

XIAO'ERGOU, China

WMO: 505480

Lat: 49.1980N

Lon: 123.7220E

Elev: 288

StdP: 97.91

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.5	(c) -33.1	(d) -39.0	(e) 0.1	(f) -35.4	(g) -36.6	(h) 0.1	(i) -32.8	(j) 6.6	(k) -16.0	(l) 5.7	(m) -15.2	(n) 0.4	(o) 20	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 31.6	(d) 20.1	(e) 29.6	(f) 19.7	(g) 27.9	(h) 19.2	(i) 22.7	(j) 27.7	(k) 21.7	(l) 26.6	(m) 20.8	(n) 25.6	(o) 2.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	16.4	25.4	20.1	15.3	24.3	19.1	14.4	23.4	68.9	27.7	65.0	26.6	61.7	25.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.3	(c) 5.3	(e) -38.0	(f) 35.1	(g) 3.0	(h) 2.0	(i) -40.2	(j) 36.5	(k) -42.0	(l) 37.7	(m) -43.7	(n) 38.8	(o) -45.9	(p) 40.3		
(a) 7.4	(b) 6.3	(c) 5.3	(e) -38.1	(f) 24.5	(g) 3.0	(h) 1.3	(i) -40.3	(j) 25.4	(k) -42.1	(l) 26.2	(m) -43.8	(n) 27.0	(o) -46.0	(p) 28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-22.0	-17.4	-6.8	4.1	12.2	18.9	21.4	18.9	12.1	2.6	-10.3	-20.5		
	DBStd	16.04	4.53	5.23	6.34	4.42	3.96	3.35	2.44	3.01	4.08	4.59	6.51	4.76		
	HDD10.0	4297	991	768	521	183	21	0	0	1	24	232	610	946		
	HDD18.3	6458	1249	1002	779	426	196	32	4	28	191	487	860	1204		
	CDD10.0	1082	0	0	0	7	88	267	354	277	86	3	0	0		
	CDD18.3	203	0	0	0	0	4	49	99	47	3	0	0	0		
	CDH23.3	2102	0	0	0	9	131	644	843	419	57	0	0	0		
	CDH26.7	655	0	0	0	2	34	241	269	100	9	0	0	0		
	WSAvg	1.8	1.0	1.2	2.0	2.7	2.7	2.0	1.8	1.6	1.8	2.0	1.4	1.0		
	PrecAvg	490	2	2	5	22	36	80	154	111	58	16	5	4		
(15) Precipitation	PrecMax	789	6	14	14	66	105	145	363	229	128	54	23	13		
	PrecMin	293	0	0	0	1	2	18	33	36	6	0	0	0		
	PrecStd	101	2	3	4	17	26	37	70	49	36	14	4	3		
	PrecStd	101	2	3	4	17	26	37	70	49	36	14	4	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.6	1.3	12.6	24.5	30.3	34.7	33.9	31.7	27.9	20.2	8.7	-4.7		
		MCWB	-9.0	-3.0	4.5	10.9	15.5	19.5	21.7	20.6	18.1	10.7	2.8	-6.4		
	2%	DB	-10.0	-2.6	9.2	20.0	26.9	31.8	31.6	29.2	24.7	16.3	4.8	-7.8		
		MCWB	-11.7	-5.8	2.5	9.0	14.3	19.1	21.3	21.0	16.2	8.6	0.5	-9.3		
	5%	DB	-12.0	-5.4	6.2	16.9	24.1	29.4	29.6	27.5	22.7	13.7	1.3	-10.3		
		MCWB	-13.3	-7.8	0.6	7.3	12.7	18.1	20.9	20.1	14.8	6.8	-1.9	-11.5		
	10%	DB	-14.0	-7.6	3.6	13.6	21.5	27.2	27.8	25.8	20.5	10.9	-1.2	-12.7		
		MCWB	-15.1	-9.5	-0.9	5.4	11.4	17.3	20.4	19.2	13.9	5.0	-3.4	-13.8		
	0.4%	WB	-8.9	-2.8	5.0	11.7	17.5	21.8	24.5	24.1	20.1	12.0	3.4	-6.3		
		MCDB	-6.6	1.3	12.2	23.4	27.7	30.0	29.8	28.9	23.4	19.0	7.9	-4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-11.8	-5.6	2.7	9.3	15.1	20.6	23.1	22.3	18.1	9.3	0.5	-9.2		
		MCDB	-10.3	-2.8	8.4	18.1	24.8	28.7	28.4	26.8	22.1	15.1	4.5	-7.8		
	5%	WB	-13.3	-7.7	0.8	7.6	13.7	19.4	22.2	21.2	16.5	7.3	-1.5	-11.5		
		MCDB	-12.0	-5.6	5.6	15.9	21.2	26.2	27.0	25.4	21.0	12.2	1.1	-10.2		
	10%	WB	-15.1	-9.5	-0.7	5.8	12.4	18.4	21.3	20.2	14.9	5.8	-3.3	-13.7		
		MCDB	-14.1	-7.6	3.3	12.8	19.5	25.0	26.0	24.4	18.9	10.3	-1.3	-12.7		
	5% DB	MDBR	14.3	16.1	14.8	13.9	14.3	12.6	10.9	11.6	14.0	13.0	12.2	12.6		
		MCDBR	15.1	17.2	16.0	20.0	19.6	17.8	15.1	14.7	17.5	17.3	14.3	13.3		
		MCWBR	13.6	14.4	10.2	11.0	9.0	6.8	6.4	7.2	9.1	10.1	10.3	11.8		
		MCWBR	14.9	16.9	15.3	18.7	16.4	14.1	11.8	11.6	13.6	15.0	13.8	13.0		
(35) Mean Daily Temperature Range	5% WB	MCWBR	13.4	14.2	9.7	10.4	8.1	5.8	6.1	6.5	7.8	9.3	10.1	11.6		
		MCWBR	13.4	14.2	9.7	10.4	8.1	5.8	6.1	6.5	7.8	9.3	10.1	11.6		
	Clear-Sky Solar Irradiance	taub	0.272	0.296	0.354	0.446	0.517	0.508	0.479	0.433	0.377	0.353	0.302	0.280		
		taud	2.253	2.204	2.114	1.967	1.847	1.981	2.119	2.226	2.319	2.270	2.286	2.205		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	780	844	843	795	753	765	782	803	816	764	730	701		
		Edh at Noon	72	98	128	166	196	173	148	127	105	92	71	67		
(44) All-Sky Solar Radiation	RadStd	RadAvg	1.80	2.98	4.42	5.13	5.56	5.80	5.38	4.89	4.05	2.84	1.79	1.39		
		RadStd	0.05	0.10	0.23	0.37	0.44	0.64	0.54	0.46	0.32	0.18	0.11	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page